

Loss Functions for Active and Passive Diffusion: Comparison with CLD and the $\tau \rightarrow 0$ Limit

1 Setup

We consider two coupled stochastic variables: image pixels x and active noise η , evolving under the forward SDEs

$$dx = (-kx + \eta) dt + \sqrt{2T_p} dW_x, \quad (1)$$

$$d\eta = -\frac{\eta}{\tau} dt + \sqrt{\frac{2T_a}{\tau^2}} dW_\eta, \quad (2)$$

with $T_p = 10^{-3}$, $T_a = 1$, and persistence time τ . The passive baseline is the single-variable OU process

$$dx = -kx dt + \sqrt{2T_e} dW_x. \quad (3)$$

2 Marginal Distribution and the $\tau \rightarrow 0$ Limit

The joint process (x_t, η_t) given (x_0, η_0) is Gaussian with means

$$\mu_x = e^{-kt} x_0 + b \eta_0, \quad \mu_\eta = e^{-t/\tau} \eta_0, \quad (4)$$

where $b = (e^{-t/\tau} - e^{-kt})/(k - 1/\tau) \rightarrow 0$ as $\tau \rightarrow 0$, and covariance matrix $\Sigma = \begin{bmatrix} M_{11} & M_{12} \\ M_{12} & M_{22} \end{bmatrix}$ with

$$M_{11} \xrightarrow{\tau \rightarrow 0} \frac{T_p + T_a}{k} (1 - e^{-2kt}), \quad M_{12} \xrightarrow{\tau \rightarrow 0} T_a, \quad M_{22} = \frac{T_a}{\tau} \rightarrow \infty. \quad (5)$$

The marginal variance of x_t therefore tends to $(T_p + T_a)/k (1 - e^{-2kt})$, which matches the passive model with $T_e = T_p + T_a$. In particular, when $T_p \rightarrow 0$ and $T_a = T_e$, the marginal distribution of x_t is identical to the passive model with temperature T_e .

3 Score-Matching Loss Functions

3.1 Passive loss

The passive model uses true score $\nabla_x \log p_t(x_t) = -(x_t - \mu_x)/M_{11}$ where $M_{11} = (T_e/k)(1 - e^{-2kt})$. The loss is implemented via `mse_loss` with two variants depending on GPU mode:

$$\mathcal{L}_{\text{passive}}^{\text{AMP}} = M_{11} \cdot (F_x - \nabla_x \log p_t)^2, \quad (6)$$

$$\mathcal{L}_{\text{passive}}^{\text{non-AMP}} = M_{11}^2 \cdot (F_x - \nabla_x \log p_t)^2. \quad (7)$$

These arise from `mse_loss`(σF_x , $\sigma \nabla_x \log p_t$) and `mse_loss`($\sigma^2 F_x$, $\sigma^2 \nabla_x \log p_t$) respectively, with $\sigma^2 = M_{11}$.

CLD passive (VPSDE). CLD’s passive baseline uses variance-preserving noise schedule $\beta(t)$, with $M_{11} = 1 - e^{-\int_0^t \beta(s) ds}$ and loss

$$\mathcal{L}_{\text{CLD-passive}} = \frac{1}{2}\beta(t) \cdot M_{11} \cdot (F_x - \nabla_x \log p_t)^2. \quad (8)$$

This follows from $(\text{score} / \text{noise_multiplier} - \varepsilon)^2 \times \text{multiplier}$ with $\text{noise_multiplier} = -1/\sigma$ and $\text{multiplier} = \beta(t)/2$.

Comparison. Equations (6) and (8) have the same M_{11} weighting; the only difference is the extra $\frac{1}{2}\beta(t)$ factor in CLD, which reflects its time-varying noise schedule (in the user’s model β is constant, so this is just a fixed prefactor absorbed into the learning rate). Equation (7) carries an additional M_{11} factor; since $M_{11} \in (0, T_e/k]$ is bounded and $\mathcal{O}(1)$, this changes the relative emphasis across diffusion times but not the minimizer.

3.2 Active loss (joint score matching)

The active model learns the full joint score $\nabla_{(x,\eta)} \log p_t(x_t, \eta_t) = -\Sigma^{-1}(x_t - \mu_x, \eta_t - \mu_\eta)^\top$:

$$\mathcal{L}_{\text{active}} = T_p \cdot \det \Sigma \cdot (F_x - \nabla_x \log p_t)^2 + T_a \cdot \det \Sigma \cdot (F_\eta - \nabla_\eta \log p_t)^2, \quad (9)$$

where

$$\nabla_x \log p_t = \frac{-M_{22}(x_t - \mu_x) + M_{12}(\eta_t - \mu_\eta)}{\det \Sigma}, \quad (10)$$

$$\nabla_\eta \log p_t = \frac{-M_{11}(\eta_t - \mu_\eta) + M_{12}(x_t - \mu_x)}{\det \Sigma}. \quad (11)$$

In practice, the loss is implemented by introducing $\mathbf{F}_x = \sqrt{\det \Sigma} F_x$ and $\mathbf{F}_\eta = \sqrt{\det \Sigma} F_\eta$, so that the targets become $\mathcal{O}(1)$ quantities independent of τ .

3.3 CLD loss (conditional score matching)

Critically-Damped Langevin Diffusion [1] augments data x with a velocity variable v and trains only the conditional score $\nabla_v \log p_t(v_t | x_t)$:

$$\mathcal{L}_{\text{CLD}} = \beta(t) f \cdot \frac{\det \Sigma}{M_{11}} \cdot (F_v - \nabla_v \log p_t)^2, \quad (12)$$

where $\det \Sigma / M_{11} = \text{Var}(v_t | x_t)$ is the conditional variance of v given x . Since $\nabla_v \log p(x, v) = \nabla_v \log p(v | x)$ (as $p(x)$ does not depend on v), the joint and conditional v -scores coincide, so CLD needs no separate x -score term. The weighting $\det / M_{11}$ (conditional variance) is more natural than the joint determinant $\det$ used in (9); the ratio is the bounded, $\mathcal{O}(1)$ quantity M_{11} .

4 Behavior of the Active Loss as $\tau \rightarrow 0$

As $\tau \rightarrow 0$ with $T_p = 10^{-3}$, $T_a = 1$ fixed, we have $\det \Sigma \approx M_{11} \cdot T_a / \tau$.

x -score term.

$$T_p \cdot \det \Sigma \approx \frac{T_p T_a M_{11}}{\tau} = \frac{T_p}{\tau} \times T_a M_{11}. \quad (13)$$

Since $\nabla_x \log p_t \rightarrow$ (passive score with $T_e = T_p + T_a$), this term becomes (T_p/τ) times the passive loss—an infinitely amplified version of the passive score-matching objective. The model F_x is therefore driven strongly to the passive score for any $T_p > 0$, no matter how small.

η -score term and x -information. The target for F_η contains an x -dependent coupling:

$$\nabla_\eta \log p_t \approx -\frac{(\eta_t - \mu_\eta)}{T_a/\tau} + \underbrace{\frac{\tau(x_t - \mu_x)}{M_{11}}}_{\rightarrow 0}. \quad (14)$$

The coupling term vanishes as $\tau \rightarrow 0$, yet the loss gradient for it is

$$T_a \cdot \det \Sigma \cdot \frac{\tau}{M_{11}} \approx T_a \cdot M_{11} \cdot \frac{T_a}{\tau} \cdot \frac{\tau}{M_{11}} = T_a^2 = \mathcal{O}(1). \quad (15)$$

The $\det \Sigma \rightarrow \infty$ weighting exactly compensates the $\tau \rightarrow 0$ shrinkage of the coupling, keeping the gradient signal finite and well-conditioned at all τ .

5 Adiabatic Elimination: Active Score Reduces to Passive

The reverse-time SDE for η (running backward from T to 0) is

$$d\eta_{\text{rev}} = \left[\frac{\eta}{\tau} + \frac{2T_a}{\tau^2} \nabla_\eta \log p_t \right] ds + \sqrt{\frac{2T_a}{\tau^2}} d\bar{W}. \quad (16)$$

Setting the mean drift to zero defines the quasi-stationary point η^* :

$$\frac{\eta^*}{\tau} + \frac{2T_a}{\tau^2} \nabla_\eta \log p_t(x, \eta^*) = 0 \implies \nabla_\eta \log p_t(x, \eta^*) = -\frac{\tau \eta^*}{2T_a}. \quad (17)$$

We now substitute the *exact* score formula (11) into (17), with $\mu_\eta \approx 0$:

$$\frac{-M_{11} \eta^* + M_{12}(x - \mu_x)}{\det \Sigma} = -\frac{\tau \eta^*}{2T_a}. \quad (18)$$

Multiplying through by $\det \Sigma$ and using $\det \Sigma = M_{11}M_{22} - M_{12}^2$ with the *exact* fluctuation-dissipation relation $M_{22} = T_a/\tau$:

$$\tau \frac{\det \Sigma}{2T_a} = \tau \frac{M_{11}M_{22} - M_{12}^2}{2T_a} = \frac{M_{11}}{2} - \frac{\tau M_{12}^2}{2T_a}. \quad (19)$$

Collecting the η^* terms gives

$$M_{12}(x - \mu_x) = \eta^* \left(M_{11} - \frac{M_{11}}{2} + \frac{\tau M_{12}^2}{2T_a} \right) = \eta^* \frac{T_a M_{11} + \tau M_{12}^2}{2T_a}, \quad (20)$$

and therefore

$$\boxed{\eta^* = \frac{2T_a M_{12}(x - \mu_x)}{T_a M_{11} + \tau M_{12}^2}}. \quad (21)$$

Closing the loop: substitute η^* into the x -score (10). With $\mu_\eta \approx 0$:

$$\nabla_x \log p_t|_{\eta=\eta^*} = \frac{-M_{22}(x - \mu_x) + M_{12} \eta^*}{\det \Sigma} = \frac{(x - \mu_x)}{\det \Sigma} \left[-M_{22} + \frac{2T_a M_{12}^2}{T_a M_{11} + \tau M_{12}^2} \right].$$

Combining the bracket over a common denominator and using $M_{11}M_{22} = \det \Sigma + M_{12}^2$:

$$\begin{aligned} -M_{22}(T_a M_{11} + \tau M_{12}^2) + 2T_a M_{12}^2 &= -T_a M_{11} M_{22} - \tau M_{22} M_{12}^2 + 2T_a M_{12}^2 \\ &= -T_a (\det \Sigma + M_{12}^2) + M_{12}^2 (\underbrace{2T_a - \tau M_{22}}_{= T_a, \text{ since } M_{22}=T_a/\tau}) \\ &= -T_a \det \Sigma. \end{aligned} \tag{22}$$

The cancellation $2T_a - \tau M_{22} = T_a$ (using $M_{22} = T_a/\tau$ *exactly*) is key. Substituting back:

$$\nabla_x \log p_t|_{\eta=\eta^*} = -\frac{T_a (x - \mu_x)}{T_a M_{11} + \tau M_{12}^2} \xrightarrow{\tau \rightarrow 0} -\frac{x - \mu_x}{M_{11}}. \tag{23}$$

Since $M_{11} \rightarrow (T_p + T_a)/k (1 - e^{-2kt}) = (T_e/k)(1 - e^{-2kt})$ as $\tau \rightarrow 0$, this is exactly the passive score for temperature $T_e = T_p + T_a$, valid for any $T_p \geq 0$ including $T_p = 0$.

Conclusion. Using the exact η -score (11) in the quasi-stationary condition (17) yields η^* (21), and substituting into (10) recovers the passive score (23) via the exact cancellation (22) (which follows from the fluctuation–dissipation relation $M_{22} = T_a/\tau$). The $\det \Sigma$ weighting in the active loss keeps the gradient for the x -coupling in F_η at $\mathcal{O}(T_a^2) = \mathcal{O}(1)$ for all τ , ensuring the network learns $\nabla_\eta \log p_t$ accurately and the passive limit is recovered.

References

- [1] T. Dockhorn, A. Vahdat, and K. Kreis, *Score-Based Generative Modeling with Critically-Damped Langevin Diffusion*, ICLR 2022.